

Twenty-fourth day of class - 11/18

Fun fact: The number with that is 109297 digits long, all of whose digits are ones, is prime. (This proof was completed in May of 2025 and uses elliptic curves!)

Last time: The baby-step-giant-step algorithm for solving the elliptic curve discrete log problem, attacks on the classical discrete log problem, and the elliptic curve factorization method.

Today: Proving that numbers are prime.

Q: Given a positive integer N , how do we test if N is prime?

A1: Enumerate all primes $p \leq \sqrt{N}$ and see if any of these $p \mid N$. If none of them do, N is prime.

The problem: This is very slow. It takes roughly $\sqrt{N}$ operations. If $N = 10^{50} + 151$, then this would take about 10^{25} steps. (Which could take a computer about $2 \cdot 10^{15}$ seconds, which is about 63 million years.)

A2: Fermat's little theorem says that if N is prime and $\gcd(a, N) = 1$, then $a^{N-1} \equiv 1 \pmod{N}$. Computing $a^{N-1} \pmod{N}$ can be done in about $\log_2(N)$ arithmetic operations. (But these operations can be kind of slow.)

So given an integer N , you can compute $a^{N-1} \pmod{N}$. If you get one, that it is quite likely that N is prime. If you don't get 1, then it is very likely N is not prime.

The problem: This test is quite fast (especially with optimized multiplication mod N), but it's not foolproof.

For example, 341 isn't prime, but $2^{340} \equiv 1 \pmod{341}$. Also, 91 isn't prime, but $3^{90} \equiv 1 \pmod{91}$. Exceptions are fairly rare though.

There are even Carmichael numbers: numbers N so that $a^{N-1} \equiv 1 \pmod{N}$ for all a with $\gcd(a, N) = 1$. There is a variant of this test though (the Miller-Rabin primality test) that if run with a random a , a composite N will pass the test with probability $\leq 1/4$. This test can be run repeatedly.

Q: Given a positive integer N , how do we prove that N is prime?

The most effective way to do this is to work in the multiplicative group mod N . The order of this group is $\phi(N) = N \prod_{p \mid N} \left(1 - \frac{1}{p}\right)$. If $N \geq 2$, then $\phi(N) \leq N - 1$ and $\phi(N) = N - 1$ if and only N is prime.

Fact 1: If $a \in (\mathbb{Z}/N\mathbb{Z})^\times$ has order k , then $a^n \equiv 1 \pmod{N}$ if and only if $k \mid n$.

Proof: If $k \mid n$, then we can write $n = km$ for some integer m . Then

$$a^n \equiv a^{km} \equiv (a^k)^m \equiv 1 \pmod{N}.$$

Conversely, if $a^n \equiv 1 \pmod{N}$, write $n = km + r$ for some $0 \leq r < k$. Then

$$1 \equiv a^n \equiv a^{km+r} \equiv (a^k)^m \cdot a^r \equiv a^r \pmod{N}.$$

Since $r < k$ and we assume that k is the smallest positive exponent so that $a^k \equiv 1 \pmod{N}$, we must have that $r = 0$. Thus, $n = km$ and so $k \mid n$.

The goal in showing that N is prime is to construct a positive integer a so that the order of a in $(\mathbb{Z}/N\mathbb{Z})^\times$ is $N - 1$.

Theorem. *Suppose that N is a positive integer, and a is an integer so that $a^{N-1} \equiv 1 \pmod{N}$ but $a^{\frac{N-1}{p}} \not\equiv 1 \pmod{N}$ for all primes $p \mid N - 1$. Then N is prime.*

Proof. By the fact above, if k is the order of $a \pmod{N}$, then $k \mid N - 1$. Either $k = N - 1$ or $k < N - 1$. If $k < N - 1$, then there is a prime p that divides $N - 1$ so that k divides $\frac{N-1}{p}$. In this case, $a^{\frac{N-1}{p}} \equiv 1 \pmod{N}$, but this is a contradiction.

Thus, $k = N - 1$. Lagrange's theorem implies that $N - 1 \mid \phi(N)$ and so $N - 1 \leq \phi(N)$ which implies that $N - 1 = \phi(N)$. This forces N to be prime. $\square$

Note: If N is prime, then there always exists an a satisfying the hypotheses of the theorem. This is because $(\mathbb{Z}/N\mathbb{Z})^\times$ is a cyclic group of order $N - 1$ and so it will contain an element of order $N - 1$.

Example: Let $N = 9 \cdot 10^{905} + 1$. Let $a = 13$. Magma can quickly verify that

$$a^{N-1} \equiv 1 \pmod{N},$$

while $a^{(N-1)/2} \not\equiv 1 \pmod{N}$, $a^{(N-1)/3} \not\equiv 1 \pmod{N}$, and $a^{(N-1)/5} \not\equiv 1 \pmod{N}$. This proves that N is prime.

This test works well, and 3 of the top 10 largest known primes can be proven to be prime using it. However, it requires us to know the prime factorization of $N - 1$, and this could be quite hard to find.

There is a variant of this test that relies on the prime factorization of $N + 1$, and all very large known prime numbers are ones for which some of the prime factors of $N - 1$ or $N + 1$ are known.

Note: The website <http://t5k.org> tracks the 5000 largest known prime numbers. The largest currently known prime is $2^{136279841} - 1$ which has about 41 million digits. All primes in the top 11 largest known primes all have more than 10 million digits.

Q: If N is a large number and the prime factorization of $N - 1$ or $N + 1$ is not known, how can we prove that N is prime?

A: Use elliptic curves!

Theorem (Goldwasser-Kilian). *Let N be a positive integer and E be an elliptic curve defined over $\mathbb{Z}/N\mathbb{Z}$. Suppose that P is a point in $E(\mathbb{Z}/N\mathbb{Z})$ and m is a positive integer. If*

- *There is a prime $q \mid m$ with $q > (\sqrt[4]{N} + 1)^2$, and*
- *$mP = (0 : 1 : 0) \pmod{N}$, and*
- *$(m/q)P$ is defined and not equal to $(0 : 1 : 0) \pmod{N}$,*

How do you pick the elliptic curve and points?

If the number N can be written in the form $a^2 + db^2$ with a , b and d , then there is a comparatively straightforward way (using elliptic curves with complex multiplication) of finding elliptic curves where the number of points mod N is $N + 1 + 2a$ and $N + 1 - 2a$.

The idea is to run through small values of d and see if we can find a number of the form $N + 1 \pm 2a$ that has a largish prime factor. Then construct the corresponding elliptic curve $y^2 = x^3 + ax + b \bmod N$.

Once you have done that, choose a random $x \bmod N$ and check to see if $x^3 + ax + b$ is a square mod N and find a square root y of $x^3 + ax + b \bmod N$. (There are fairly quick.) Then let $P = (x, y)$.

The exact running time is hard to give a rigorous upper bound to, but there is theoretical evidence that the algorithm should run in time of around $c(\log N)^4$ and is amenable to parallelization.

Record computations

On May 26, 2025, Andreas Enge (author of “fastECP”) and Paul Underwood announced that they had proven that $\frac{10^{109297}-1}{9}$ is prime.

I wasn’t able to find many details about that calculation, but they had earlier shown that $\frac{10^{86453}-1}{9}$ is prime. In their calculation, they use 2979 different q ’s. The total computation time was about 383 CPU years (less than 4 months of real time was needed). The details are stored in a way that they can be verified much more quickly (but the verification still takes 190 CPU days).